

Notes: Barrage (RES, 2020)

Optimal Dynamic Carbon Taxes in a Climate–Economy Model with Distortionary Fiscal Policy

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1 Big picture

- **Research question.** How should carbon emissions be taxed when governments must also raise revenue using distortionary taxes?
- **Benchmark being challenged.** In many climate–economy models with lump-sum taxes, the optimal carbon tax is Pigouvian.
- **Main friction.** Real governments rely on distortionary labour, capital, and consumption taxes. Carbon taxes raise direct revenue but can shrink other tax bases.
- **Core theoretical insight.** Carbon emissions are disinvestment in environmental capital. Underpricing emissions distorts intertemporal investment incentives, like capital income taxation.
- **Key theorem-level result.** If capital income taxes are optimally zero, the optimal carbon tax fully internalizes production damages at the Pigouvian rate.
- **Quantitative contribution.** COMET combines a DICE-style climate model with distortionary fiscal instruments and a production/utility damage split.
- **Main numerical result.** Distortionary taxes lower optimal carbon tax schedules by 8–24% relative to lump-sum finance.
- **Welfare magnitude.** Even after this adjustment, optimal carbon taxation produces roughly \$21–26 trillion in welfare gains.

2 Data and calibration

2.1 Fiscal data and baseline taxes

The calibration uses IMF Government Finance Statistics and effective tax rate estimates for 107 countries. Government spending is split between government consumption and transfers. Baseline effective tax rates are approximately 33.4% for capital income, 28.1% for labour income, and 11.1% for consumption.

Interpretation: The calibration makes carbon taxation a genuine fiscal instrument rather than an environmental wedge added to a frictionless economy.

2.2 Climate damages: production versus utility

Production damages include agriculture, vulnerable markets, and sea-level rise. Utility damages include amenity value, ecosystems, and resettlement. Health and catastrophic damages are mixed.

Interpretation: This split is central because production damages affect future output and investment incentives, while utility damages enter welfare directly and are discounted through the marginal cost of public funds.

2.3 Climate–economy model

The carbon-cycle and temperature modules follow DICE. Emissions enter atmospheric, upper-ocean/biosphere, and deep-ocean reservoirs; atmospheric carbon affects radiative forcing and temperature; production uses labour, capital, and energy.

Interpretation: COMET adds fiscal structure to integrated assessment modelling, allowing the value of climate damages to depend on the tax system.

3 Models and theoretical specifications

3.1 Household optimality

$$\frac{U_{ct}}{U_{c,t+1}} = \beta (1 + r_{t+1}(1 - \tau_{k,t+1}) - \delta), \quad -\frac{U_{lt}}{U_{ct}} = w_t(1 - \tau_{l,t}).$$

Interpretation: Capital taxes distort saving and labour taxes distort labour supply. Carbon taxes matter because emissions alter future output and utility.

3.2 Government budget constraint

$$G_t^C + G_t^T + B_t^G = \tau_{l,t}w_tL_t + \tau_{I,t}E_t + \tau_{E,t}E_t^M + \tau_{k,t}(r_t - \delta)K_t + \rho_t B_{t+1}^G.$$

Interpretation: The carbon tax is one instrument in a broader public finance system.

3.3 Pigouvian carbon tax decomposition

$$\tau_E^{Pigou,T} = \tau_E^{Pigou,U} + \tau_E^{Pigou,Y}.$$

The utility component captures marginal utility damages; the production component captures marginal output losses.

Interpretation: The Pigouvian levy is scenario-dependent because it is evaluated at the allocation generated by the fiscal environment.

3.4 Carbon–capital tax link

$$\tau_k^* = 0 \quad \Rightarrow \quad \tau_E^* \text{ internalizes production damages at the Pigouvian rate.}$$

Interpretation: Underpricing emissions is like taxing investment in the climate asset: agents emit too much today and leave too little environmental capital for future production.

4 Identification and quantitative design

The exercise is a calibrated structural counterfactual. The key comparison is across model economies that differ in fiscal instruments and carbon tax schedules. The credibility of the quantitative result rests on the calibration and on the theoretical decomposition of fiscal interactions rather than on exogenous policy variation.

5 Tables and figures with detailed interpretation

5.1 Table 1: Climate damage categorization and Table 2: Government expenditure shares

Read: Table 1 classifies climate impacts into production, utility, and mixed channels. Table 2 calibrates government spending to 31.07% of GDP, split between government consumption and social benefits.

Interpretation: Table 1 determines which climate damages enter the production side of the model and therefore which damages are connected to the carbon-capital tax logic. Table 2 anchors the fiscal problem.

Economic message: Climate policy and public finance are linked because climate damages affect both welfare and production while governments finance large expenditure obligations.

What not to over-read: The classification is a modelling discipline, not a claim that every real-world climate impact can be neatly assigned to one category.

TABLE 1 Climate damage categorization	
Impact category	Classification
Agriculture	Production
Other vulnerable markets (energy services, forestry, etc.)	Production
Sea-level rise coastal impacts	Production
Amenity value	Utility
Ecosystems	Utility
Human (re)settlement	Utility
Catastrophic damages	Mixed
Health	Mixed

Table 1: Climate damage categorization.

TABLE 2 Government expenditure shares (2005)		
	% of GDP	% of government spending
Government consumption	17.75	57
Social benefits	13.32	43
Total	31.07	

Data sources: IMF Government Finance Statistics, IMF International Financial Statistics.

Table 2: Government expenditure shares, 2005.

5.2 Table 3: Summary of key calibration parameters

Read: Table 3 reports preference, production, damage, abatement, and government parameters.

Interpretation: The table shows how COMET is built from DICE-style climate dynamics, GHKT-style production, and fiscal calibration. The recalibration of utility parameters keeps baseline labour supply comparable across fiscal settings.

Economic message: The numerical result emerges from a full calibration in which taxes, labour supply, capital accumulation, damages, and abatement costs interact.

What not to over-read: Parameters provide a benchmark for sensitivity analysis, not a definitive estimate of all climate-economy primitives.

Parameter	Value	Sources and notes
Preferences: $U(c_t, l_t, T_t) = \frac{[c_t(1-\xi_t)^\gamma]^{1-\sigma}}{1-\sigma} + \frac{(1+aq_t^\gamma)^{-(1-\sigma)}}{1-\sigma}$		
σ	1.5	Nordhaus (2010)
β	(0.985) ¹⁰	Nordhaus (2010)
ς	$\begin{cases} 2.2381 & \text{if distortionary} \\ 2.0785 & \text{if first-best} \end{cases}$	Jointly match Frisch elasticity (0.78) and rationalize base labour supply ($l_{2005} = 0.2272$)
γ	$\begin{cases} 0.7178 & \text{if distortionary} \\ 1.2985 & \text{if first-best} \end{cases}$	at base year marginal product of labour w_{2005} with $\tau_{l,2005} = 0.3609$ or $\tau_{l,2005} = 0$, respectively
aq	$\begin{cases} 0.00023808 & \text{if distortionary} \\ 0.00027942 & \text{if first-best} \end{cases}$	Match disutility from 2.5°C warming equiv. to 0.46% output loss given the relevant ς, γ
Production damages: $(1 - D_t(T_t)) = \frac{1}{1+\theta_1 T_t^2}$	θ_1	Match 1.29% output loss from 2.5°C warming
Final goods production: $F_t(A_{1t}, T_t, L_{1t}, K_{1t}, E_t) = (1 - D_t(T_t))A_{1t}(K_{1t}^\alpha L_{1t}^{1-\alpha} E_t^\nu)$		
α	0.3	GHKT (2014)
ν	0.03	GHKT (2014)
δ	0.6513	Match annual $\delta^{Tr} = 10\%$ (in % per decade)
Energy good production: $E_t = A_{2t}(K_{2t}^{1-\alpha_E} L_{2t}^{\alpha_E})$		
α_E	0.403	U.S. Bureau of Economic Analysis Data (2000–10)
Abatement costs: $\Theta_t(\mu_t, E_t) = \frac{\bar{a} p_t^{backstop}}{1+a_0 \exp(b_0 - b_1(\mu_t, E_t))} \cdot (\mu_t, E_t)$ for $t \leq 2250$, $= 0$ afterwards		
$\bar{a}$	0.9245	
a_1	$= 49.9096 + 1.0995 \log(t)$	Minimize sum of squared errors in difference to DICE abatement costs
b_0	$= 15.5724 + 3.4648 \log(t)$	
b_1	$= 13.0210 + 1.5725 \log(t)$	
$p_t^{backstop}$	$= 0.63(1 + \exp(-0.05 \cdot t))$	Nordhaus (2010), \$1000/mtC (\$2005)
Government:		
$\tau_{l,2005}$	$\begin{cases} 36.09\% & \text{if distortionary} \\ 0\% & \text{if first-best} \end{cases}$	GDP-weighted global avg. effective rate
$\tau_{k,2015}$	$\begin{cases} 33.4\% & \text{if distortionary} \\ 0\% & \text{if first-best} \end{cases}$	
G_{2005}^C	98.395 Strillion	$= (0.331)(0.57)(\text{GDP}_{2005}^{\text{World}} \cdot 10)$ (\$2005)
G_{2005}^T	73.713 Strillion	$= (0.331)(0.43)(\text{GDP}_{2005}^{\text{World}} \cdot 10)$ (\$2005)
B_0	34.659 Strillion	$= (0.6263)(\text{GDP}_{2005}^{\text{World}})$ (\$2005), World Bank WDI

Notes: See [Supplementary Appendix](#) for full parameter listing and further details.

Table 3: Summary of key calibration parameters.

5.3 Table 4: Main results and Figure 1: Optimal carbon tax paths across fiscal scenarios

Read: Table 4 summarizes the main policy scenarios; Figure 1 shows that first-best carbon taxes are highest throughout the century, while distortionary-tax paths are lower.

Interpretation: Distortionary fiscal policy lowers the optimal carbon tax schedule relative to the lump-sum benchmark, but the optimized schedule remains large and rising.

Economic message: The right comparison is which carbon tax schedule is optimal in a fiscal system with costly revenue-raising.

What not to over-read: Lower taxes under distortionary finance do not mean climate policy is undesirable; the welfare gains remain large.

Fiscal scenario	Carbon and energy	τ_1^* Avg. 2025-2255	MCF	Carbon tax τ_2^*		T_1 Max	ΔC_{2015} (Strl.)	$\Delta \text{Welfare}^b$ % ΔC_1 τ_1
				2025 \$/mC	% Adjust. from FB (6)			
Labour								
Capital								
Variable	None (until 2115) ^a	38.3%	1.06	0	0	4.31	\$0.05	0.00
$\tau = 38.3\%$	Variable	None (until 2115)	34.3%	1.43	0	4.30	\$22.42	0.82
Optimized	Variable	None (until 2115)	2.8%	1.06	0	4.31	\$24.88	0.90
$\tau = 38.3\%$	Variable	"Wrong" (First-best)	31.9%	1.37	71	105	\$26.21	0.95
$\tau = 38.3\%$	Variable	Optimized ^d	31.1%	1.35	49	77	\$21.27	0.78
Variable	Variable	"Wrong" (First-best)	37.9%	1.05	71	105	\$22.03	0.80
Variable	Variable	Optimized ^d	37.9%	1.06	58	91	\$45.77	1.63
Optimized	First-best	0	0	1.00	71	105	\$96.81 ^f	3.25

adjustment of optimal vs. first-best carbon tax during twenty-first century. ^aRelative to BAU scenario (1a). Equivalent variation change in aggregate initial consumption r permanent change in consumption % ΔC_1 τ_1 . ^bCarbon taxes are allowed after 2115 so as to keep the analysis in an appropriate range for the (smooth) damage function. of high initial tax followed by $\sim 0\%$ tax (see [Supplementary Appendix](#) for graph of τ_2^* over time). ^cIncludes non-zero intermediate input tax on clean and dirty energy $\tau_{intC=eq}$ in 2025). See [Supplementary Appendix](#) for derivation and alt. results with $\tau_{intC=eq}$. ^dIncludes non-zero intermediate input tax on clean and dirty energy ($\tau_2^* = eq$ in 2025). See [Supplementary Appendix](#) for derivation and alt. results with $\tau_{intC=eq}$. ^eCalculation uses utility parameters from second-best model to compare both first-id-best allocations. However, these are not strictly comparable as leisure preferences are actually calibrated to $\tau_{10}=0$ in the first-best but $\tau_{10}=36.09\%$ in the second-best.

Table 4: Main results.

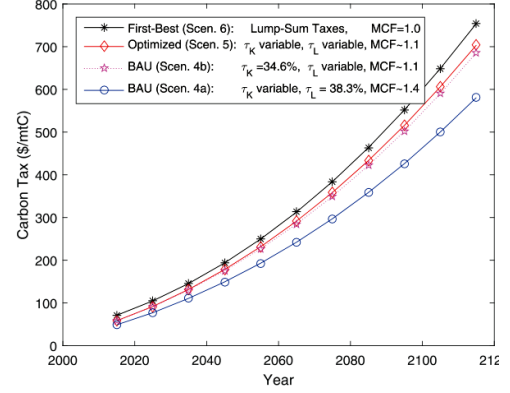


Figure 1: Optimal carbon tax paths across fiscal scenarios.

5.4 Figure 2: Optimal versus Pigouvian carbon tax levels and Table 5: Optimal versus Pigouvian decomposition, 2025

Read: Distortionary taxes lower the Pigouvian level and may push the optimal carbon price below that Pigouvian level. Table 5 decomposes the 2025 carbon price into utility, production, and fiscal-constraint terms.

Interpretation: There are two fiscal effects: a smaller economy has smaller marginal damages, and optimal policy no longer equals Pigouvian policy because carbon taxes interact with income-tax constraints.

Economic message: The optimal-Pigouvian gap is not a single tax-interaction penalty; it is the sum of several fiscal and damage-channel components.

What not to over-read: The decomposition is for 2025 and for particular fiscal constraints; the components change over time.

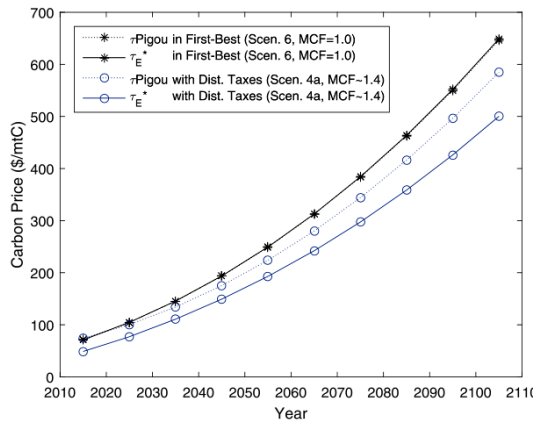


Figure 2: Optimal versus Pigouvian carbon tax levels.

Scenario 4b ^a : Variable labour tax, fixed capital income tax $\tau_L=34.57\%$ (MCF ~ 1.1)				
	Utility damages	Production damages	Fiscal constraint interactions	Total
τ_{E1}^{Pigou}	$\Sigma \beta^j \frac{U_{T+1}}{U_{11}}$	$\Sigma \beta^j \frac{U_{T+1}}{U_{11}} F_{T+1}$		
	\$22	\$82		\$104
τ_{E1}^*	$\Sigma \beta^j \frac{U_{T+1}}{U_{11}} \frac{1}{MCF_1}$	$\Sigma \beta^j \frac{\lambda_{T+1}}{\lambda_{11}} F_{T+1}$	$\Sigma \beta^j \frac{\Psi_{T+1}}{\lambda_{11}} \frac{\partial F_{T+1}}{\partial \tau_{L+1}} (1 - \tau_L)$	
	\$18	\$69	\$5	\$92
Scenario 4a ^a : Fixed labour tax $\tau_L=38.25\%$, variable capital income tax (MCF ~ 1.3)				
	Utility damages	Production damages	Fiscal constraint interactions	Total
τ_{E1}^{Pigou}	$\Sigma \beta^j \frac{U_{T+1}}{U_{11}}$	$\Sigma \beta^j \frac{U_{T+1}}{U_{11}} F_{T+1}$		
	\$19	\$73		\$93
τ_{E1}^*	$\Sigma \beta^j \frac{U_{T+1}}{U_{11}} \frac{1}{MCF_1}$	$\Sigma \beta^j \frac{\lambda_{T+1}}{\lambda_{11}} F_{T+1}$	$\Sigma \beta^j \frac{\lambda_{T+1}}{\lambda_{11}} \frac{\partial F_{T+1}}{\partial \tau_{K+1}} (1 - \tau_K)$	
	\$15	\$71	-\$6	\$80

Table 5: Optimal versus Pigouvian decomposition, 2025.

5.5 Figure 3: Optimal versus Pigouvian carbon tax ratios and Figure 4: Ratios by production-damage share

Read: Figure 3 plots optimal/Pigouvian ratios over time. Figure 4 shows the ratio rising with the share of damages that affect production.

Interpretation: Production damages behave like losses to future productive capital, so the Ramsey logic pushes the carbon tax closer to Pigouvian levels.

Economic message: Treating climate damages as pure utility losses can understate optimal carbon prices in a distortionary-tax economy.

What not to over-read: Figure 4 does not prove that all climate damages are production damages; it shows sensitivity to classification.

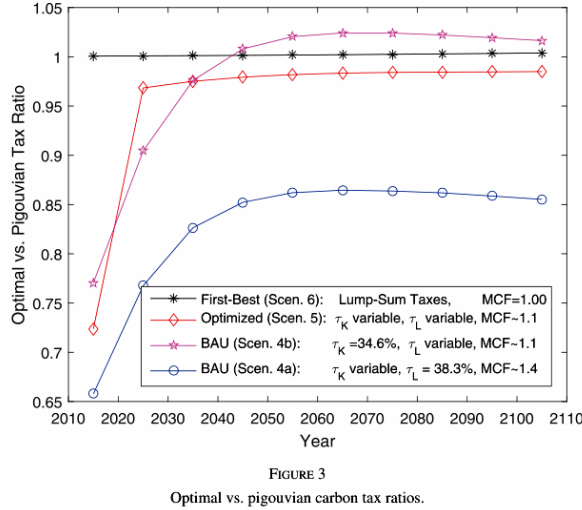


Figure 3: Optimal versus Pigouvian carbon tax ratios.

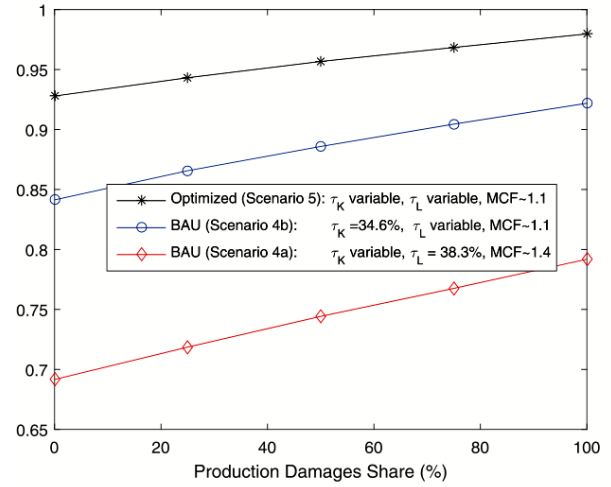


Figure 4: Ratios by production-damage share.

5.6 Table 6: Sensitivity analysis—policy and Table 7: Sensitivity analysis—welfare

Read: Tables 6 and 7 vary Frisch elasticity, consumption elasticity, and government spending. Optimal carbon taxes remain below first-best with distortionary taxes, while welfare gains remain large.

Interpretation: The main result is robust: distortionary fiscal environments call for lower carbon taxes than first-best, but the welfare case for carbon taxation remains large.

Economic message: Fiscal conditions matter quantitatively. A first-best SCC should not be mechanically imported into a high-MCF tax system.

What not to over-read: Robustness does not mean numerical invariance. The direction is stable, but magnitudes depend on preferences, spending needs, and fiscal constraints.

	τ_l^*	τ_k^*	MCF	Carbon tax τ_{FB}^*			T_l max
				2015 \$/mtC	2025 \$/mtC	%Adjustment from FB (6.)	
				Frisch elasticity = 2			
6. First-best	0	0	1	71	104	—	2.97
5. Optimized distortionary	43.7%	4.2%	1.13	53	85	−15%	3.04
4a. BAU labour tax $\bar{\tau}_l = 39\%$		40.9%	1.75	40	66	−36%	3.11
4b. BAU capital tax $\bar{\tau}_k = 40.9\%$	39.2%		1.19	50	82	−20%	3.06
Consumption elasticity: $\sigma = 2$							
6. First-best	0	0	1	49	71	—	3.16
5. Optimized distortionary	45.5%	3.9%	1.02	38	59	−14%	3.2
4a. BAU labour tax $\bar{\tau}_l = 40.75\%$		39.4%	1.4	30	47	−34%	3.20
4b. BAU capital tax $\bar{\tau}_k = 39.4\%$	40.9%		1.03	36	58	−18%	3.20
Consumption elasticity $\sigma = 1.1$							
6. First-best	0	0	1	104	151	—	2.69
5. Optimized distortionary	38.9%	2.0%	1.06	90	139	−5%	2.71
4a. BAU labour tax $\bar{\tau}_l = 35\%$		33.2%	1.45	73	113	−23%	2.75
4b. BAU capital tax $\bar{\tau}_k = 33.2\%$	35.2%		1.06	90	138	−6%	2.71
Government spending −10%							
6. First-best	0	0	1	72	106	—	2.95
5. Optimized distortionary	35.9%	2.1%	1.04	64	99	−3%	2.96
4a. BAU labour tax $\bar{\tau}_l = 32\%$		29.6%	1.30	53	84	−19%	2.97
4b. BAU capital tax $\bar{\tau}_k = 29.6\%$	32.2%		1.03	63	99	−4%	2.97
Government spending +10%							
6. First-best	0	0	1	70	103	—	3.00
5. Optimized distortionary	49%	3.8%	1.10	52	83	−16%	3.05
4a. BAU labour tax $\bar{\tau}_l = 45\%$		38%	1.64	41	66	−35%	3.13
4b. BAU capital tax $\bar{\tau}_k = 38\%$	45.2%		1.13	51	81	−19%	3.07

Notes: All figures represent average of 2015–115 values unless noted otherwise.

Table 6: Sensitivity analysis—policy.

	Tax scenario		Δ Welfare	
	Labour	Capital	Carbon and energy	ΔC_{2015} (\$tril.) % ΔC_l yr
Frisch elasticity = 2				
1a.	39.6%	$\bar{\tau}_k = 40.9\%$	None (until 2115)	—
3b.	39.3%	$\bar{\tau}_k = 40.9\%$	“Wrong” (First-Best)	\$19.3 0.72%
4b.	39.2%	$\bar{\tau}_k = 40.9\%$	Optimized	\$22.4 0.83%
3a.	$\bar{\tau}_l = 39\%$	42.4%	“Wrong” (First-Best)	\$15.1 0.57%
4a.	$\bar{\tau}_l = 39\%$	40.9%	Optimized	\$21.4 0.79%
5.		Optimized		\$72.6 2.54%
Consumption elasticity: $\sigma = 2$				
1a.	41.2%	$\bar{\tau}_k = 39.4\%$	None (until 2115)	—
3b.	40.9%	$\bar{\tau}_k = 39.4\%$	“Wrong” (First-Best)	\$9.0 0.43%
4b.	40.9%	$\bar{\tau}_k = 39.4\%$	Optimized	\$10.1 0.48%
3a.	$\bar{\tau}_l = 40.75\%$	40.2%	“Wrong” (First-Best)	\$7.6 0.36%
4a.	$\bar{\tau}_l = 40.75\%$	39.4%	Optimized	\$9.5 0.45%
5.		Optimized		\$37.4 1.71%
Consumption elasticity: $\sigma = 1.1$				
1a.	35.7%	$\bar{\tau}_k = 33.2\%$	None (until 2115)	—
3b.	35.2%	$\bar{\tau}_k = 33.2\%$	“Wrong” (First-Best)	\$44.6 1.17%
4b.	35.2%	$\bar{\tau}_k = 33.2\%$	Optimized	\$45.5 1.19%
3a.	$\bar{\tau}_l = 35\%$	34.6%	“Wrong” (First-Best)	\$39.5 1.04%
4a.	$\bar{\tau}_l = 35\%$	33.2%	Optimized	\$44.2 1.16%
5.		Optimized		\$73.9 1.89%
Government spending −10%				
1a.	32.5%	$\bar{\tau}_k = 29.6\%$	None (until 2115)	—
3b.	32.1%	$\bar{\tau}_k = 29.6\%$	“Wrong” (First-Best)	\$23.1 0.79%
4b.	32.2%	$\bar{\tau}_k = 29.6\%$	Optimized	\$23.2 0.79%
3a.	$\bar{\tau}_l = 32\%$	29.6%	“Wrong” (First-Best)	\$22.1 0.76%
4a.	$\bar{\tau}_l = 32\%$	29.6%	Optimized	\$22.7 0.77%
5.		Optimized		\$38.5 1.29%
Government spending +10%				
1a.	45.6%	$\bar{\tau}_k = 38\%$	None (until 2115)	—
3b.	45.6%	$\bar{\tau}_k = 38\%$	“Wrong” (First-Best)	\$11.2 0.45%
4b.	45.2%	$\bar{\tau}_k = 38\%$	Optimized	\$20.8 0.83%
3a.	$\bar{\tau}_l = 45\%$	40.2%	“Wrong” (First-Best)	\$11.2 0.45%
4a.	$\bar{\tau}_l = 45\%$	38.2%	Optimized	\$19.6 0.79%
5.		Optimized		\$56.3 2.16%

Table 7: Sensitivity analysis—welfare.

6 Short summary

- The paper studies optimal carbon taxes as part of the fiscal system.
- The theoretical contribution is a link between carbon taxation and capital income taxation.
- If capital taxes are optimally zero, carbon taxes should fully internalize production damages.
- With constrained income taxes, optimal carbon taxes may deviate from Pigouvian rates.
- COMET distinguishes production damages from utility damages.
- Optimal carbon tax paths are 8–24% lower with distortionary taxes than with lump-sum taxes.
- Welfare gains from carbon pricing are about \$21–26 trillion in the benchmark.
- Sensitivity exercises confirm that fiscal adjustments remain important.

7 Conclusion

7.1 Core conclusion

Barrage shows that optimal carbon pricing cannot be separated from fiscal policy. Carbon taxes interact with the same intertemporal margins that make capital taxation costly.

7.2 Why the Pigouvian benchmark changes

The Pigouvian benchmark is useful but not automatically optimal in a second-best fiscal system. With distortionary taxes, the carbon tax must account for marginal cost of funds and tax-base effects.

7.3 Production-damage channel

Utility damages are like a climate public good. Production damages raise the value of future output when avoided, making climate policy resemble investment policy.

7.4 Policy implication

Carbon tax recommendations should specify the fiscal environment. Revenue recycling is not secondary; it changes the marginal cost of public funds and thus the optimal tax.

7.5 Limits and future research

The paper abstracts from uncertainty, distributional politics, incomplete commitment, and heterogeneous countries in the benchmark. Future work can integrate fiscal shocks, stochastic climate risks, and richer tax systems.

7.6 Final takeaway

The right lesson is not “tax carbon less because taxes are distortionary.” It is “tax carbon with the fiscal system in mind.”